The adjoint $SU(n)$ sigma-MGF

Proof and exact checks for $n = 2, 3, 4$

Computational proof companion

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Abstract

We derive the compact Molien–Weyl formula for the adjoint representation of $SU(n)$ and test the manageable cases $n = 2, 3, 4$. Exact Schur-character integration confirms the quadratic Killing form, the alternating bracket tensor, and the symmetric cubic precisely when $n \geq 3$.

1 Formula

The adjoint weights of $SU(n)$ are zero with multiplicity $n-1$ and $e_i - e_j$ for $i \neq j$. The general Haar/Cauchy proof yields

$$\boxed{\mathcal{S}_{SU(n),\mathrm{Ad}} = \frac{1}{n!}\,\mathrm{CT}_z \left[D_{A_{n-1}}(z) \prod_a (1-t_a)^{-(n-1)} \prod_{i\neq j} (1 - t_a z_i/z_j)^{-1} \right]}.$$

Indeed, expanding the determinant gives $\sum_\lambda s_\lambda(\mathbf{t})\chi_{\mathbb{S}_\lambda \mathfrak{sl}_n}$; Haar integration retains exactly the invariant multiplicity.

2 Low-degree structure

The Killing form supplies h_2, and $\kappa(X,[Y,Z])$ supplies e_3 for every $n \geq 2$. For $n \geq 3$ there is also the symmetric invariant

$$(X, Y, Z) \longmapsto \mathrm{Tr}(XYZ + XZY),$$

which supplies h_3. For $SU(2)$ it vanishes. Thus, through degree three,

$$\mathcal{S}_{SU(2),\mathrm{Ad}} = 1 + h_2 + e_3 + O_{\geq 4},$$

and, for $n \geq 3$,

$$\mathcal{S}_{SU(n),\mathrm{Ad}} = 1 + h_2 + h_3 + e_3 + O_{\geq 4}.$$

Chevalley restriction gives $\mathcal{S}(t, 0, \ldots) = \prod_{d=2}^{n} (1-t^d)^{-1}$.

3 Direct and exact verification

For each $n = 2, 3, 4$, the notebook generates the A_{n-1} roots, appends the correct zero weights, and constructs the diagonal adjoint torus matrix. It compares the determinant with the product over eigenvalues. It then computes every partition of degrees zero through three by Jacobi–Trudi and applies the exact Weyl functional

$$\frac{1}{|W|}\,\mathrm{CT}(Df) = \mathrm{CT}\left(f \prod_{\alpha>0} (1 - z^{-\alpha}) \right).$$

n	$\dim \mathfrak{sl}_n$	$[h_2]$	$[h_3]$	$[e_3]$	all other $d \leq 3$
2	3	1	0	1	0
3	8	1	1	1	0
4	15	1	1	1	0

All 21 partition-by-case comparisons pass. The tested representation dimensions $3, 8, 15$ are large enough to distinguish the exceptional $n = 2$ behavior from the stable $n \geq 3$ cubic while remaining directly inspectable.

Reproducibility. Run `su_adjoint_verification.py` with marimo.